University Bulk Order & Predictive Procurement Analytics

Machine-learning driven demand forecasting and risk segmentation for university textbook procurement.

TOTAL PREDICTED DEMAND	TOTAL PROJECTED SPEND	DIGITAL VS PHYSICAL	HIGH-RISK OPT-OUT RAT
3,985 Units	$324,069.3 M	28% / 72%	4%
	Millions equivalent	Share of predicted units	Share of records with Opt-Ou

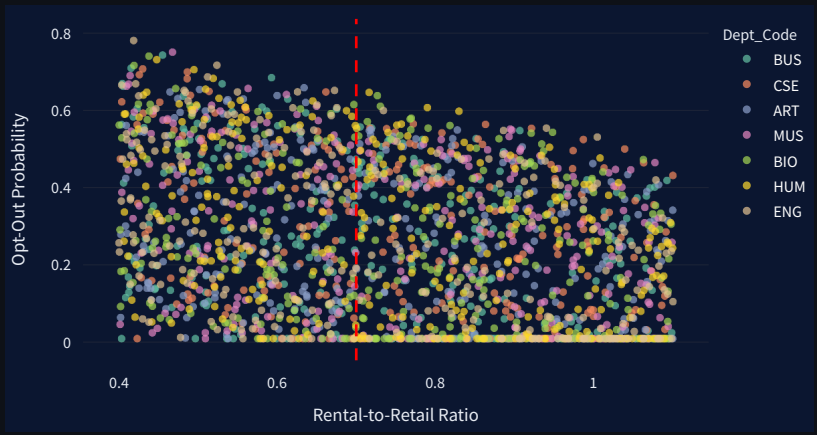
Feature Engineering Pipeline & Key Indicators

Price Sensitivity & Opt-Out Threshold

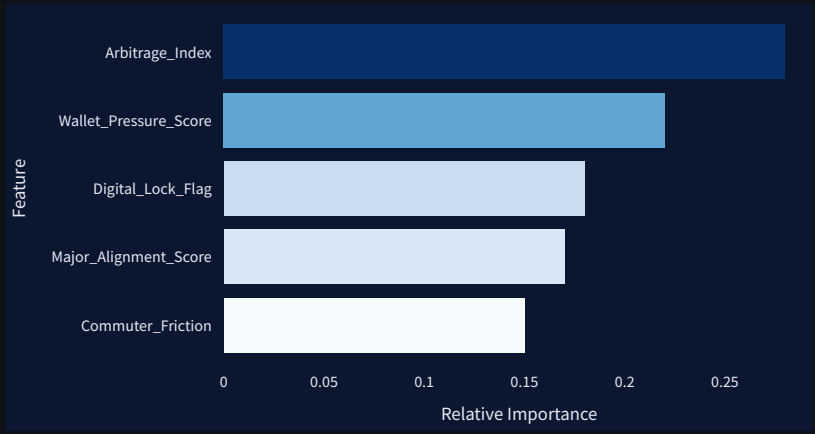
Rental-to-Retail Ratio Opt-Out Threshold

0.70

0.40 1.00



Feature Importance (Model Explainability)



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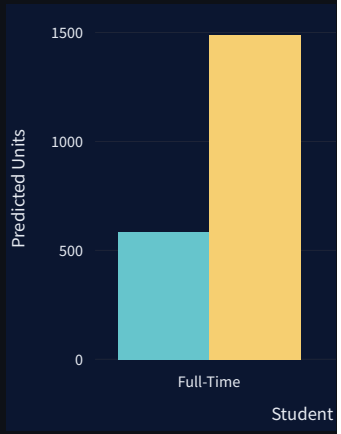


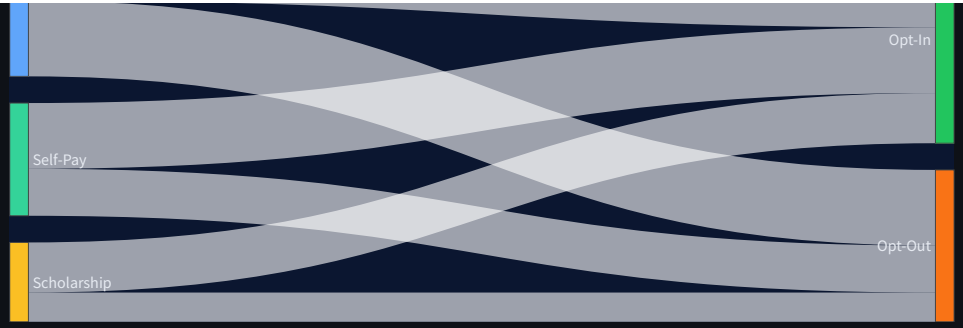
Procurement Planning & Strategy

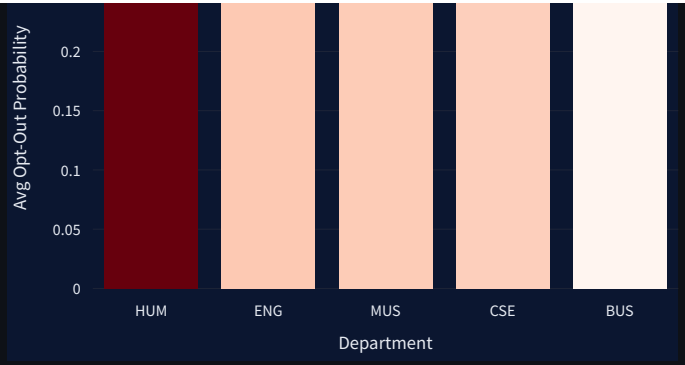
Funding Source Planning & Strategy

Procurement Risk: Top 5 High-Risk Departments

Recommended Actions

- Prioritize **bundle price negotiatio** departments: MUS, BUS, ENG.





- Increase **digital access options** fo time student segments.
- Review **required vs recommende** to-retail arbitrage.
- Share **early adoption reports** wit realistic order quantities.